Procedural-fairness compute pipelines

Process Consistency

How stable is the prediction under semantic noise?

- Trained model f + test sample x (sample_n = 500)
- Gaussian noise on numerics σ   {0.1, 0.3, 1.0, 3.0}
- JS divergence between original and perturbed predictions. Mean over rows; bootstrap CI

Voice / Representation

How much does the model use modifiable features?

- Trained model f + test sample x
- TreeExplainer SHAP attribution (or KernelExplainer)
- Voice = modifiable attribution share over total attribution. Enrichment adjusts for feature count

Model-Flippability

Can the prediction be reversed by a sparse counterfactual?

- Trained model f + test sample x (sample_n = 30)
- Greedy k_max = 1 counterfactual over numeric features within ±1σ
- Sparsity: 1 – #changed/|F|
Validity: x′ flipping ŷ

Explanation-Actionability

Does the counterfactual flip a modifiable feature?

- Greedy k_max = 1 counterfactual reuses model-flippability output
- Is the flipped feature in F_mod?
- Fraction over rows; bootstrap CI

Foundation: F = F_mod   F_imm, the modifiable / immutable feature partition (ascribed-vs-achieved framing of Newman 2020).